

Ethics, Law & Responsible Data Collection

Week 2 · Foundations module · Textbook Chapter 2

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Before we collect a single row, we settle the harder question — not *can* we, but **should** we, and if so, how responsibly?

Monday | Concepts & notebook intro

- The legal landscape, `robots.txt`, and a five-question ethical framework

Wednesday | Notebook lab

- `robots.txt`, User-Agent headers, rate limiting — and building `responsible_get()`

Friday | Textbook revisions

- GitHub, hands-on — and everyone files their first issue against Chapter 2

Companion notebook

`ch-02-ethics.ipynb`

Bring a laptop

Wednesday is hands-on — we send live requests and watch what the server sees.

1. Monday • Concepts

Monday at a glance

00:00 – 00:04	The ethics of retrieval
00:04 – 00:20	The law: the CFAA, four rulings, and what is still moving
00:20 – 00:26	Terms of Service & privacy policies
00:26 – 00:34	Activity: audit a platform you use
00:34 – 00:44	Discussion: four dispatches from the scraping wars
00:44 – 00:50	The framework, and Wednesday

Two things to have open

A browser, and the four articles from the reading. Half of today is you reading primary sources — not me talking.

“Web scraping” is a popular phrase — and a faintly pejorative one. Data is **valuable**: other people invested time, money, and effort to create and host what you are about to retrieve programmatically.

So the question is never simply “**can I scrape this?**” but “**should I — and if so, how do I do it responsibly?**”

This chapter builds a **decision framework** to revisit before every collection project in this course and beyond. Three layers of guidance, none sufficient alone: **law**, **technical norms** (`robots.txt`), and **ethics**.

One idea to carry all semester

“It’s technically possible” is not the same as “it’s legal,” and “it’s legal” is not the same as “it’s ethical.”

The CFAA: a 1986 law meets the modern web

The **Computer Fraud and Abuse Act** was enacted in 1986, expanding a narrower 1984 statute, during a national panic about “computer crime.” The web did not exist.

Its key provision prohibits “**unauthorized access**” to a computer system. What the statute never does is say what makes access to a *public web page* unauthorized.

That gap matters because the CFAA is a **criminal** statute. The question “was that authorized?” is not an abstraction — it is the difference between a research method and a federal charge.

Forty years of litigation have been an argument about that one word.

Why you should care

Every collection decision you make this semester sits underneath a criminal statute drafted before the web existed — and interpreted by courts that are still disagreeing.

Two events that shaped everything after

Aaron Swartz (2011–2013). Swartz used MIT’s network to bulk-download academic articles from JSTOR. Federal prosecutors charged him under the CFAA with eleven counts. Together they carried decades of theoretical exposure. The prosecutors argued that a violation of a terms-of-use agreement was “unauthorized access.” He died by suicide in 2013, at 26.

Cambridge Analytica (2018). A researcher collected Facebook data through the platform’s API for ostensibly academic purposes, then handed it to a political consulting firm. The fallout closed APIs across the industry — for everyone, including the researchers who had done nothing wrong.

One shock each way

Swartz taught researchers that a vague statute could be aimed at them.

Cambridge Analytica taught platforms that open access was a liability.

Together they explain why the doors were closing before AI ever showed up (Ch. 3).

The facts. A police sergeant ran a license-plate search in a database he was *authorized* to use — but did it for a bribe, not police work. He was charged under the CFAA’s “exceeds authorized access” clause.

The holding. The Supreme Court adopted a “**gates-up-or-down**” reading: the CFAA targets entering areas you are not entitled to enter. It does *not* reach misusing information you were allowed to see.

Why it matters to you. Before *Van Buren*, a prosecutor could plausibly argue that breaking a website’s stated terms was a federal crime. After it, that theory is very hard to sustain.

The test, in one line

Did you go **around a gate**, or did you break a **rule** about data you could already reach?

Only the first is the CFAA’s business.

This is the case people cite *wrong*. It has three acts:

1. hiQ, an analytics company, scraped **public** LinkedIn profiles. LinkedIn sent a cease-and-desist and blocked it.
2. The Ninth Circuit held — in 2019, and **again in 2022** after the Supreme Court sent it back in light of *Van Buren* — that scraping publicly accessible data likely does **not** violate the CFAA. No authentication gate was bypassed.
3. Later in 2022, hiQ **lost** on LinkedIn's **breach-of-contract** claim: it kept scraping after agreeing to the ToS and receiving the C&D. The case settled with hiQ agreeing to stop.

The half-quote

You will hear “*hiQ* means scraping public data is legal.”

That is act two of three. The company still lost, and stopped scraping.

The CFAA is not the only law in the room.

The facts. Researchers wanted to run **audit studies** of algorithmic discrimination. They planned to make test profiles on housing and employment sites. The profiles would show if the algorithms treated people differently by race. This method violates the Terms of Service of those sites. The researchers feared prosecution under the CFAA, so they sued *first*.

The holding. A federal district court held in 2020 that merely violating a website's Terms of Service is **not** a CFAA crime.

Why it matters. Audit studies are how we know what we know about algorithmic discrimination. This ruling is modest — one district court — but it is real cover for a whole research method.

The tension it names

Platforms write the rules that govern how they may be studied.

A ToS that forbids test accounts is also a ToS that forbids auditing you for discrimination.

Four rulings, one through-line

Case	Holding	What it means for you
<i>Van Buren</i> (2021)	CFAA reaches gates, not use restrictions	Breaking a rule $\neq$ a federal crime
<i>hiQ</i> (2019, 2022)	Public scraping likely not CFAA — but breach of contract	Winning on the CFAA is not winning
<i>Sandvig</i> (2020)	ToS violation alone is not a CFAA crime	Audit research has some cover
<i>Bright Data</i> (2024)	Logged-out public scraping survived contract claims	Never agreeing means no contract

The through-line: the CFAA polices **authentication barriers**, not **use restrictions** — gates, not rules. But what left criminal law did not disappear. It moved to **contract**, which changes which court you are in and what the penalty is, not whether you are exposed.

Two rulings against the same data broker mark where the line is now:

- **Meta v. Bright Data** (early 2024) — Meta's breach-of-contract claims **failed**. The scraping at issue collected only public data *without logging in*, so the account terms Meta relied on never attached.
- **X Corp. v. Bright Data** (later 2024) — **dismissed**, with the court warning against letting platforms use contract law to claim exclusive ownership of public web data.

These are trial-court rulings, not nationwide rules. But the direction is consistent with *hiQ*: risk concentrates at **authentication barriers and contractual relationships**, not at public visibility.

The distinction to internalize

Logged out — you agreed to nothing, so there is no contract to breach.

Logged in — you agreed, and the ToS is now enforceable against you.

Same data. Very different exposure.

The New York Times v. OpenAI and Microsoft (filed 2023) asks a question that no earlier case asked. Many similar suits ask it too. Is it infringement to collect copyrighted content and **train a model** on it? Or is it fair use?

That question is unresolved. But the *litigation* has already changed the web, without waiting for a ruling:

- publishers hardened their sites against crawlers
- AI-specific directives appeared in `robots.txt`
- exclusive licensing deals carved up access

You are downstream

You are not a party to this fight, but you live with its result.

A door closed against OpenAI is closed against your class project too. We will meet this again as **enclosure** in Ch. 3.

State privacy law reaches you

The U.S. has no comprehensive federal privacy law, so states filled the gap. California wrote the **CCPA/CPRA** first. By 2026 more than a dozen states have comprehensive statutes. They include the **Colorado Privacy Act**, which covers data about Colorado residents.

They import GDPR-like concepts: **personal data rights**, **purpose limitation**, and obligations on whoever processes the data.

The **GDPR** itself reaches EU residents' personal data regardless of where *you* are sitting.

If your scraped dataset contains personal information about residents of these places, these laws can apply to **you** — student status is not an exemption.

The habit this should build

Treat personal data as a **liability, not an asset**. Every extra field is extra risk you are choosing to hold. Collect the minimum that answers your question.

Follow the through-line to its conclusion: as the CFAA retreated, **contract advanced**. The Terms of Service is where the real constraint on your work now lives.

Every major platform's ToS restricts automated collection. When you created an account, you agreed. Consequences run from account suspension to IP blocking to litigation.

And the agreement is **asymmetric**: you accepted it once, by clicking. They amend it by posting. Tan (2024) documents platforms quietly rewriting terms and privacy policies to permit **AI training** on data users had already shared.

This course's own warning

Code in several chapters retrieves data in ways that may violate a platform's ToS. It is here to teach technique.

You are responsible for the terms of anything you access. When in doubt: use the API, or don't collect it.

Reading a ToS or privacy policy

These documents are long, and they are written not to be read.
So do not read them linearly — **search them like a scraper.**

Keywords worth a Ctrl-F:

access | script | scrape | crawl | automated | bot | API | research | data |
abuse | rate | machine learning

Four questions to come out with:

1. Is **automated access** addressed at all?
2. Is there a **research or academic** exception?
3. What does it say about **personal data** and **AI training**?
4. When was it **last updated** — and what changed?

Question 4 is the sharp one

Tan (2024) found changes as small as a few words — inserting “artificial intelligence” or “machine learning” into a policy you agreed to years ago.

In February 2024 the **FTC** warned that rewriting a privacy policy to retroactively cover data already collected may be “**unfair or deceptive.**”

Activity: audit a platform you use

Eight minutes. Select a platform that you use. Open its Terms of Service or its privacy policy.

1. Search the document for the keywords on the last slide.
2. Answer the four questions.
3. Find one clause that surprised you.

Then report two things. First, does this platform permit the data collection that you plan for this class? Second, how sure are you?

Public institutions have terms too

The Colorado General Assembly publishes every bill, sponsor, and vote — and still has a privacy policy governing its site.

“Public data” never means “no terms.” It is the first place students assume otherwise.

Discussion: four dispatches from the scraping wars

Ten minutes. Four minutes in groups, then six minutes for reports. Each group takes one article:

1. **The terms move** — Tan (2024), on ToS rewritten to permit AI training
2. **The web pushes back** — Koebler (2024a), on measurable mass blocking
3. **The defense misfires** — Koebler (2024b), on sites blocking the wrong bots
4. **The rules get ignored** — Mehrotra et al. (2024), on scraping around a block

Three questions each

1. What **changed**, concretely?
2. Who **gains** and who is **harmed**?
3. What does it mean for **your** project this semester?

What the four add up to

The informal bargain that made the open web scrapable is coming apart — in four moves:

1. **The contract layer shifts under you.** Terms are quietly amended to permit AI training on data already shared — until the FTC warns that doing it retroactively may be “unfair or deceptive” (Tan 2024).
2. **The norm layer hardens.** Of 14,000 sites studied, full AI restrictions went from ~1% to 5–7% in a year — and 28% of the most actively maintained, critical sources added restrictions (Koebler 2024a).
3. **The norm layer does not actually work.** Reuters and Condé Nast blocked two *retired* Anthropic crawlers while the active one walked through. Meanwhile iFixit was hit ~1M times in a day, and Read the Docs ate \$5,000 in bandwidth (Koebler 2024b).
4. **Some actors ignore it entirely.** Perplexity reached blocked pages from an unpublicized IP — and then fabricated a claim that a named police officer had committed a crime (Mehrotra et al. 2024).

Where the bill lands

Blanket blocks cannot tell a bad-faith AI crawler from a search engine, an archive, or a student project. **The enforcement aimed at them catches you.**

An ethical decision framework

Before any project, work through five questions:

1. **Legality** — CFAA, GDPR, local law, the site's ToS?
2. **Consent** — did people consent? Is it truly public, or shared with a reasonable expectation of limited visibility (Zimmer 2010)?
3. **Proportionality** — only the data you need? Could a smaller or less sensitive set answer the question?
4. **Privacy** — any PII? Could individuals be re-identified? What if it leaks?
5. **Server impact** — a meaningful load? Rate-limited? Scraping off-peak?

No framework substitutes for judgment — but working these questions surfaces the risks.

IRBs & web data

Scraping becomes *human-subjects research* when you collect data about **identifiable people** to produce **generalizable knowledge**. Public officials' votes: usually not. Social-media profiles for health or politics: almost certainly. When in doubt, ask your IRB.

Wednesday: `robots.txt` in the wild

Today was the **law** and the **contract**. Wednesday is the third layer — the **technical norm** — and we go straight at it with code.

You will get the `robots.txt` file of many real sites and compare them. Then you will build the `responsible_get()` function. You will use that function for the rest of the semester.

Come with a prediction. Pick two sites you would expect to differ — a nonprofit and a newspaper, say — and guess which restricts more, and why. We will check.

Bring a laptop

We send live requests and watch what the server sees. Your `webdata` environment should already work — if it does not, fix it **before** Wednesday.

2. Wednesday · Notebook Lab

A `robots.txt` at a site's root declares what automated agents may fetch. Read it raw, or parse it and ask permission per URL:

```
import requests
from urllib.robotparser import RobotFileParser

print(requests.get("https://en.wikipedia.org/robots.txt").text[:400])
# User-agent: * / Disallow: /w/ / Allow: ... / Crawl-delay: ...

rp = RobotFileParser()
rp.set_url("https://en.wikipedia.org/robots.txt")
rp.read()

rp.can_fetch("*", "https://en.wikipedia.org/wiki/Python_(programming_language)") # True
rp.can_fetch("*", "https://en.wikipedia.org/w/api.php") # False
rp.crawl_delay("*") # seconds between requests, or None -> pick a polite default
```

The nuance: `/wiki/` articles are open, but `/w/` (dynamic pages + API) is disallowed *for crawlers*. Deliberate API clients follow different norms (Ch. 10) — `robots.txt` is one input, not the whole rulebook.

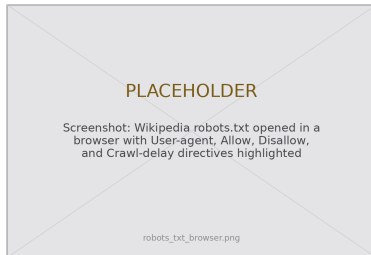
What robots.txt reveals across platforms

```
sites = {"Wikipedia": "https://en.wikipedia.org/robots.txt",
        "Reddit": "https://www.reddit.com/robots.txt",
        "NY Times": "https://www.nytimes.com/robots.txt"}

for name, url in sites.items():
    lines = requests.get(url).text.strip().split("\n")
    disallow = sum(1.startswith("Disallow") for l in lines)
    allow = sum(1.startswith("Allow") for l in lines)
    print(f"{name}: {len(lines)} lines, {disallow} Disallow, {allow} Allow")
```

Wikipedia — permissive where it matters (openness is the mission; data flows through its API + dumps). **Reddit** — tolerates access within limits. **NY Times** — restrictive; the content is a commercial product. Reading it tells you *what kind of organization* you're dealing with.

Two cautions. First, it is a **norm, not a law**: a violation is not illegal, but it is impolite and the site can block your IP address. Second, a **404** means *no rules*. It does not mean “crawl aggressively.”



Activity: survey a sector

Three sites tell you little. A **sector** tells you about a business model. Select one row. Get the `robots.txt` file of each site in that row. Compare them:

- **Search** — `google.com`, `bing.com`, `duckduckgo.com`
- **Social** — `facebook.com`, `instagram.com`, `linkedin.com`, `x.com`, `reddit.com`
- **Commerce** — `amazon.com`, `ebay.com`, `ticketmaster.com`
- **Streaming** — `netflix.com`, `hulu.com`, `spotify.com`
- **News** — `nytimes.com`, `washingtonpost.com`, `wsj.com`, `theguardian.com`
- **Public** — `leg.colorado.gov`, `census.gov`, `sec.gov`

Report back

- Most and least restrictive in your row?
- What does the pattern say about **how each makes money**?
- Does the **public** row look different — and should it?

Which AI crawlers does this site block?

Since 2023, `robots.txt` has grown a second job: naming **AI training crawlers**. Search each file you fetched for:

GPTBot (OpenAI) | ClaudeBot (Anthropic) | CCBot (Common Crawl) |

Google-Extended | Applebot-Extended | Meta-ExternalAgent |

PerplexityBot

Then look for the failure Koebler (2024b) found: a site blocking a **retired** crawler name while the operator's **active** one walks straight through. Copy-pasted blocklists rot.

The deeper limit: `robots.txt` has no enforcement mechanism. It is a sign on a door, not a lock — Mehrotra et al. (2024) documents a company reaching blocked pages from an unpublicized IP address.

So why respect it?

Because you are not trying to get away with something — you are trying to do defensible research.

The norm is weak against bad actors and completely effective against your reputation.

Identify yourself, then slow down

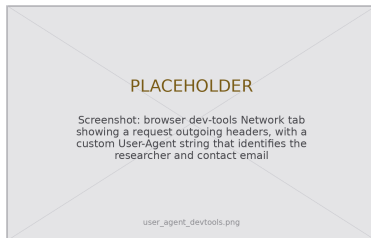
Two habits make you a good citizen of the web — a truthful `User-Agent` and rate limiting:

```
import time, requests

headers = {
    "User-Agent": "WebDataScience/1.0 (INFO 4617; brian.keegan@colorado.edu)"
}

for url in urls:
    response = requests.get(url, headers=headers)
    # ... process the response ...
    time.sleep(1) # a good default: about one request per second
```

The default `requests` `User-Agent` marks you as an anonymous bot. A good one **identifies you and gives contact info**, so operators can reach you instead of simply blocking you — and some APIs *require* it. Hundreds of requests per second is an accidental **denial-of-service**: respect any `Crawl-delay` and stay well inside documented API limits.



responsible_get(): build it once

Custom User-Agent, rate limiting, and error handling appear in *every* scraping project. Wrap them into one reusable function instead of re-implementing them ad hoc:

```
import requests, time

def responsible_get(url, headers=None, delay=1.0):
    """GET with responsible defaults: identify, rate-limit, fail loudly."""
    default_headers = {
        "User-Agent": "WebDataScience/1.0 (INFO 4617; you@colorado.edu)"
    }
    if headers:
        default_headers.update(headers)
    try:
        response = requests.get(url, headers=default_headers, timeout=30)
        response.raise_for_status() # treat 4xx/5xx as failures, not data
        time.sleep(delay) # rate-limit by default
        return response
    except requests.exceptions.RequestException as e:
        print(f"Request failed ({type(e).__name__}): {url}")
        return None
```

Why it's built that way — and reused everywhere

Every default encodes a principle

- **Custom User-Agent** — identify & give contact
- **delay defaults ON** — aggressive scraping must be a *conscious* override, not an oversight
- **raise_for_status()** — a 403/404 fails loudly instead of feeding your parser garbage
- **broad except** — `ConnectionError`, `Timeout`, `SSLError` log rather than crash the run
- **timeout=30** — never hang forever on a dead server
- **pass a site's Crawl-delay as delay** to honor its stated preference

You'll import this again

`responsible_get()` returns in **Ch. 6** (scraping HTML tables), **Ch. 7** (Wayback snapshots), and throughout the **API chapters**.

Building responsible defaults into your *tools* is more reliable than remembering to add them each time.

Lab: work these, then show-and-tell

Work in pairs. Do these exercises from the notebook:

1. Compare the `robots.txt` files of five sites in one category. Report the patterns that you find.
2. Read one Terms of Service. Find three clauses that limit automated collection. Report if research is an exception.
3. Apply the five-question framework to this plan: collect dating-app profiles to study self-presentation.
4. Measure five requests with `time.time()`. Use a 1-second delay, then a 2-second delay. Compare the times with your intent.
5. Select a site that you use each day. Compare its `robots.txt`, its Terms of Service, and the framework. Write 300 words. Show where they disagree.
6. 5617: read Fiesler et al. (2020), the EFF page about the CFAA, and *Van Buren*. Then write a two-page memo about collection from platforms

Show-and-Tell

Bring one of these to class:

- a bug that you could not solve
- data that you collected and find interesting
- a question for a research design

“Can I scrape this?” is the wrong question.

Ask instead: *should* I — and if so, how do I do it responsibly?

Legality · Consent · Proportionality · Privacy · Server impact.

3. Friday • Textbook Revisions

Friday: your first contribution

Week 1 was a single short session and we ran out of room before anyone filed anything. So today we do it properly, and slowly.

By the end of class you will have:

1. a working GitHub setup, verified with me in the room
2. **one real issue filed** against Chapter 2

Pull requests start **next week**. Today is the cheaper, faster half of contributing — and it is the half that finds the problems.

00:00 – 00:06	The four objects
00:06 – 00:18	Setup, together
00:18 – 00:26	What to look for
00:26 – 00:32	Anatomy of an issue
00:32 – 00:47	File it
00:47 – 00:50	Next week

Four objects you'll use all semester

- **Repository** — the project: every file and its full history.
- **Issue** — a written report: “here is a problem.” Costs nothing but a browser.
- **Pull request** — a proposed change: “here is the fix, side by side with the original.”
- **Review** — a response to a PR: comment line-by-line, then approve or request changes.

Issue vs. pull request

An **issue** says *something is wrong*. A **pull request** says *here is the correction* — and shows the exact diff.

Issues are cheap and fast; PRs are the real contribution. **Today: an issue.** From next week on: pull requests.

Setup, together: two paths

To file an issue you need only a browser. That is today's task, and it works for everyone.

We will also set up your **tools** now, while I am here to help. Next week you need them for pull requests. Select one tool:

GitHub Desktop — download it, sign in, then **Clone** the textbook repository. Each later step is a button: branch, commit, push, Create Pull Request.

The **gh CLI** — install it, run `gh auth login`, then use these commands:

- `gh repo clone cuinfoscience/Web-Data-Science-Book`
- `gh issue create` (yes, you can file today's issue this way)

Checkpoint — raise a hand

You are done when Desktop shows the textbook repository on your computer, or when `gh repo clone` completes with no error.

Tell me if a step fails. Today is the one class period when I can repair it for you.

High-value targets — pick one and scope it to a single PR:

- **A confirmed gap.** Wikipedia’s live `robots.txt` now has a `Disallow: /trap/` rule the chapter doesn’t mention — a honeypot for bad bots. Explain what it’s for and add it to the comparison.
- **Close a gap in “Common Issues to Debug.”** Not covered yet: a blocked page can return **HTTP 200 with a Cloudflare challenge page**, not a 403 — `raise_for_status()` won’t catch it. Add this as a worked case.
- **Add a figure — there are currently zero in this chapter.** A clean five-question framework diagram, an annotated `robots.txt` anatomy, or a CFAA case timeline.
- **Add a missing citation.** The *Meta v. Bright Data* (2024) and *X Corp.* rulings are described in prose with no source link — find and add one.
- **Strengthen an exercise.** Add a rubric or expected output to the framework exercise (#3), or a starter cell so it self-checks.

Anatomy of a good issue

Every issue you file this semester uses one skeleton:

Title: <type>: <specific thing> in Ch. 2

Location: Section, and the file
(ch-02-ethics.qmd)

Problem: What you did, expected, got.
Paste the code and output.

Why: Who this affects. One sentence.

Proposal: The concrete change you'd make.

If you cannot fill in **Location** and **Problem** with specifics, you do not have an issue yet — you have a hunch. Go back and reproduce it.

Scored out of 10

- **Located** (2) — section and file named
- **Evidenced** (3) — I can hit the same wall
- **Actionable** (3) — one concrete change
- **Reviews** (2) — from next week on

Merged is not the bar

A well-evidenced issue I decline still earns full credit. A one-character typo fix does not.

File it: your first issue

Fifteen minutes. Each student files one issue.

1. Read `ch-02-ethics.qmd` in the textbook repository. You can also read the published chapter.
2. Find one problem. Use the revision menu, or use a different problem that stopped you.
3. Do the step again and record the result. Copy the passage, the command, or the output.
4. Select the **Issues** tab. Select **New issue**. Complete the skeleton. Select **Submit**.
5. Copy the URL of your issue into the Canvas assignment.

Read the open issues first. If another student filed your issue, add a comment to it. Write what that issue does not include. This also gets credit.



One browser tab. No install required.

Next week: from issues to pull requests

An issue says *something is wrong*. Starting next week you say *here is the fix* — and back it with a diff.

A strong PR

- One focused change, clearly titled
- Explains the problem it fixes
- Builds (`quarto render`) without errors
- Matches the book's voice and notation
- Discloses any AI assistance

A strong review

- Runs / reads the change, not just skims
- Names specifics (line, claim, code)
- Separates “must fix” from “nice to have”
- Is generous — assume good faith
- Ends with a clear verdict

Next week: **The Post-API Age** — why the open web is closing, and how researchers adapt.

Key takeaways

1. “Can I?” is not “should I?” — the real question is how to collect data *responsibly*.
2. The law is unsettled: the CFAA targets **bypassing gates** (Van Buren), but **contracts and ToS** still bite (hiQ); state privacy laws and GDPR reach personal data.
3. `robots.txt` is a **norm, not a law** — read it, respect `Disallow` and `Crawl-delay`; a 404 means no rules, not open season.
4. **Identify yourself** (custom `User-Agent`) and **rate-limit** (`time.sleep`) — then wrap both into `responsible_get()` and reuse it.
5. Run the **five-question framework** before every project; when in doubt, use an API, collect less, and err toward caution.